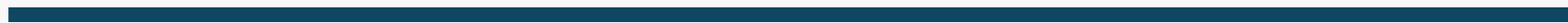




ML Regression Challenge

Ge Zeng



LASSO Delivers the Best Predictive Performance

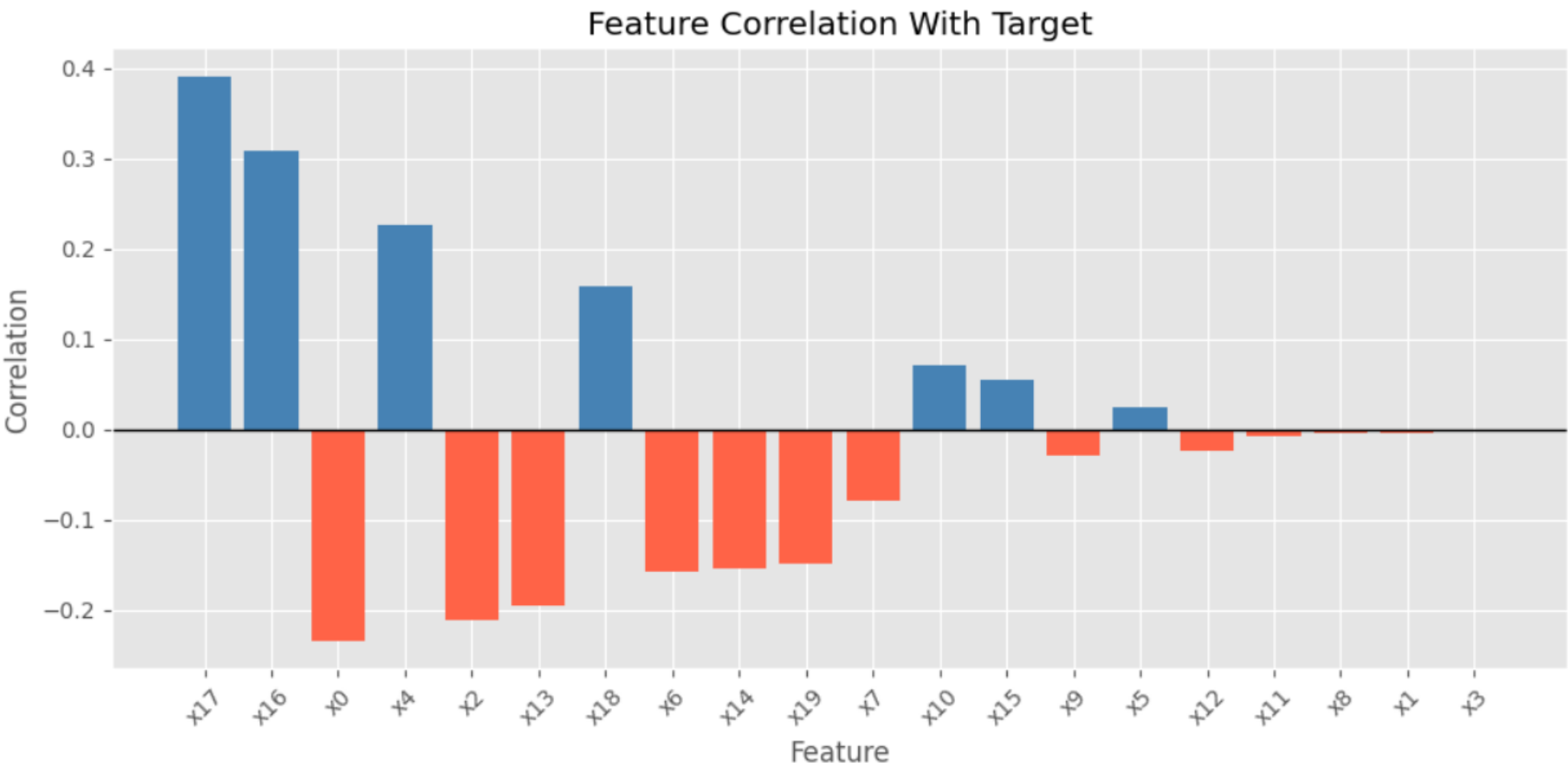
- Objective: Predict a continuous target using 20 anonymized numerical features, with model performance evaluated using R^2
- Best-performing approach: **LASSO regression with mean imputation and feature standardization**
 - L1 regularization controls model complexity by shrinking less informative feature coefficients toward zero
 - Public leaderboard: $R^2 = 0.57087$, ranking 8th
 - Private leaderboard: $R^2 = 0.54065$, ranking 2nd

Submission and Description	Private Score ⓘ	Public Score ⓘ	Selected
 summer2026_lasso_top10_predictions.csv Complete · 3d ago · Zeng_Ge	0.54562	0.56082	☐
 summer2026_lasso_no_standardize_predictions.csv Complete · 3d ago · Zeng_Ge	0.54069	0.57002	☐
 summer2026_ridge_predictions.csv Complete · 3d ago · Zeng_Ge	0.54030	0.57057	☒
 summer2026_elasticnet_predictions.csv Complete · 17d ago · Zeng_Ge	0.54006	0.56997	☐
 summer2026_lasso_predictions.csv Complete · 17d ago · Zeng_Ge	0.54065	0.57087	☒

Exploratory Analysis Reveals Strong Linear Signals

- The training dataset contains 2,500 observations, 20 numerical features, a continuous target, and an identifier
- Each feature has approximately 0.6% - 1.24% **missing values**
- Feature means are close to zero, but their **scales vary**, but their scales vary substantially, supporting **feature standardization**
- The target is also centered around zero, with a wide range from approximately $-1,215$ to 968
- x17 and x16 have the strongest correlations with the target (0.39 and 0.31)

	count	mean	std	min	25%	50%	75%	max
x0	2478.0	0.370	12.181	-41.102	-7.709	0.630	8.378	45.473
x1	2479.0	0.073	3.923	-11.863	-2.609	0.069	2.682	13.916
x2	2475.0	0.172	10.458	-44.496	-7.006	0.376	7.260	33.730
x3	2478.0	-0.011	1.868	-6.714	-1.300	-0.022	1.285	6.206
x4	2473.0	-0.120	9.757	-34.224	-6.920	0.224	6.622	27.218
x5	2469.0	-0.040	2.688	-10.246	-1.851	-0.114	1.746	8.186
x6	2476.0	0.101	14.164	-47.955	-9.780	-0.167	9.465	59.323
x7	2485.0	0.202	14.377	-55.423	-9.637	0.268	9.620	52.877
x8	2480.0	-0.035	4.290	-15.795	-2.860	-0.058	2.912	15.874
x9	2472.0	-0.066	3.651	-14.736	-2.459	-0.074	2.269	12.169
x10	2475.0	-0.237	7.766	-27.845	-5.589	-0.235	5.065	25.583
x11	2474.0	0.099	5.595	-19.144	-3.720	0.071	3.899	18.652
x12	2483.0	-0.073	9.698	-35.566	-6.756	-0.095	6.705	31.076
x13	2476.0	0.126	9.189	-29.498	-6.181	-0.071	6.565	27.818
x14	2479.0	0.008	13.779	-47.448	-9.297	-0.142	9.481	52.552
x15	2470.0	-0.038	3.547	-11.579	-2.527	-0.101	2.421	12.248
x16	2470.0	0.186	12.834	-50.306	-8.253	0.168	8.923	39.195
x17	2471.0	0.160	13.307	-40.541	-8.862	0.390	9.195	42.598
x18	2475.0	0.294	10.347	-37.318	-6.647	0.346	7.356	40.345
x19	2471.0	-0.400	14.394	-49.885	-9.708	-0.649	9.573	49.209
target	2500.0	1.777	149.001	-1215.473	-75.011	-0.101	72.747	968.431
Id	2500.0	2510.862	1438.516	3.000	1272.750	2544.500	3760.000	4999.000

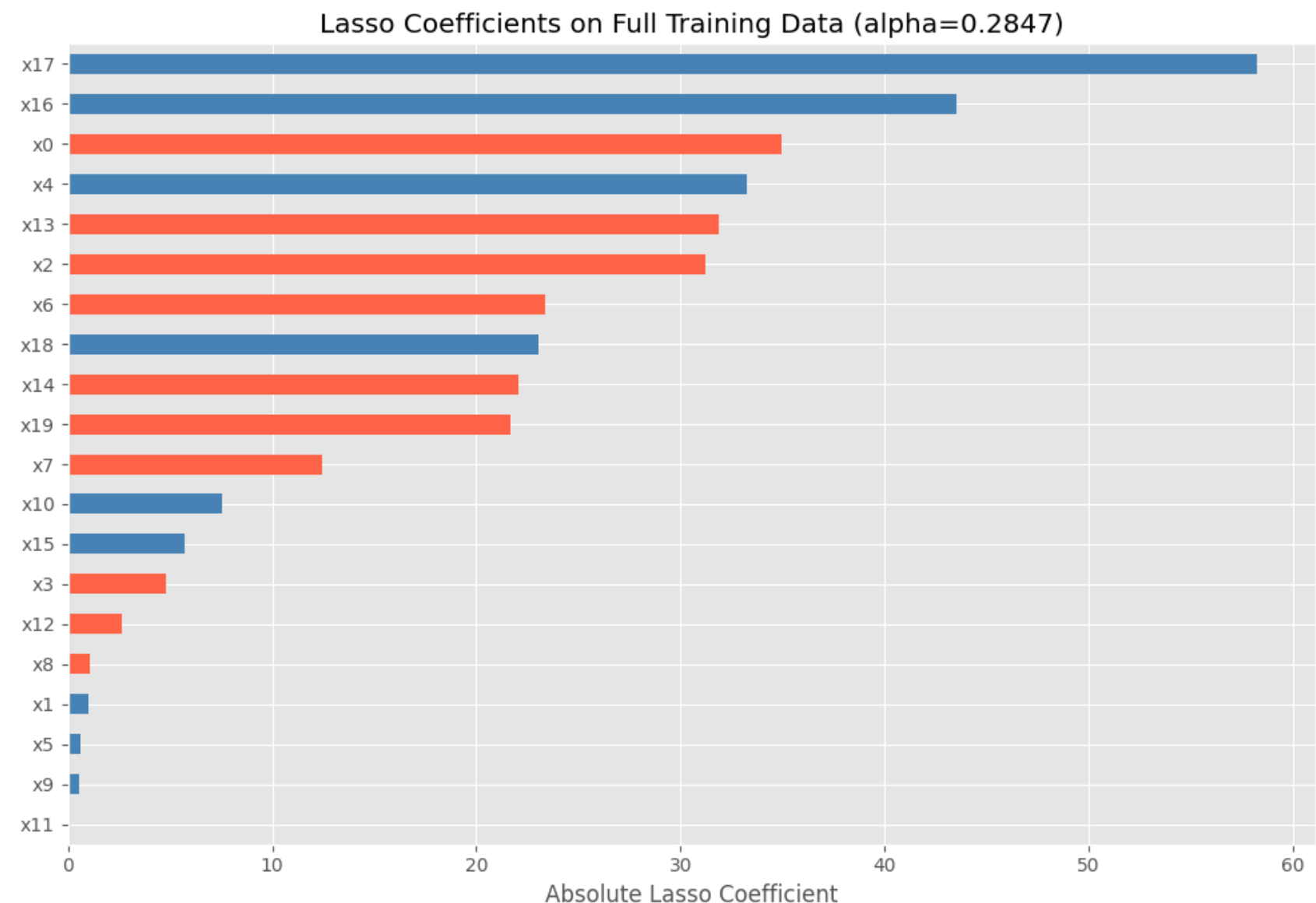


Final Model: Cross-Validated LASSO Regression

- Split the training data into 80% training-set and 20% validation-set
- Replace missing values with training-set means, then standardize features using training-set statistics
- Use 5-fold cross-validation to select the best LASSO regularization strength: $\alpha = 0.2847$
- Refit on all 2,500 training observations using the selected α
 - **R^2 on full train data: 0.5424**
- LASSO retained 19 of 20 features, shrinking the coefficient for x11 to zero
 - x17 and x16 have the largest coefficient magnitudes, consistent with the correlation observed during EDA

Lasso coefficients for each raw feature, ranked by $|\text{coefficient}|$:

x17	58.215
x16	43.557
x0	-34.968
x4	33.247
x13	-31.907
x2	-31.201
x6	-23.350
x18	23.070
x14	-22.044
x19	-21.656
x7	-12.452
x10	7.511
x15	5.709
x3	-4.783
x12	-2.651
x8	-1.053
x1	0.967
x5	0.622
x9	0.566
x11	-0.000



Other Model Selection and Finding

- **KNN:** Tested whether nearby observations could provide better predictions
 - Best configuration used 21 neighbors with uniform weighting, achieving full-training $R^2 = 0.3976$.
- **Random Forest:** Explored nonlinear relationships and feature interactions
 - Across 5-fold cross-validation, average training R^2 was 0.4238 versus validation R^2 of 0.2724, suggesting overfitting issues
- **Elastic Net:** Combined L1 and L2 regularization.
 - Best configuration used $\alpha = 0.1$ and L1 ratio = 0.8, achieving full-training $R^2 = 0.5422$
- **Ridge:** Applied L2 regularization while retaining all features
 - Selected $\alpha = 31.62$ produced full-training $R^2 = 0.5424$

These alternatives did not improve my submission results, so I retained Prediction 1 with LASSO



Thank you!

